

Module 10 Lab:

Mockito

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PART 1: Introduction & Technical Implementation

This lab demonstrated isolated unit testing in Java using Mockito with a small Java Maven project provided by the TA called mockito-lab-demo. Claude.ai was used to extract the code from the PDF to avoid any copy-paste errors. This exercise was conducted using Terminal – confirming installation of Java JDK, Apache Maven; IntelliJ IDEA was previously installed but not necessary for this demonstration (see Appendix 1, Figure 1).

Mocking is a way to test functionality of a class in isolation (Mockito, n.d.). This is something that my projects have utilized in the past to test a system before the API or databases are ready, we still need to be able to ensure that our part of the code is working properly.

The project's pom.xml includes three test-scoped dependencies that make the entire Mockito/JUnit setup work. JUnit Jupiter is the JUnit 5 testing framework -- it provides the `@Test` annotation, the assertion methods, and the test lifecycle that Maven Surefire runs. `mockito-junit-jupiter` integrates Mockito with JUnit 5 so that annotations like `@Mock`, `@InjectMocks`, and `@ExtendWith(MockitoExtension.class)` work automatically without manual setup. Finally, the `maven-surefire-plugin` is the engine that actually executes the tests when we run `mvn test` or `mvn clean test`. Surefire also generates the TXT, XML, and HTML reports that appear under `target/surefire-reports` and `target/site`. All three dependencies are marked with test scope, meaning they are only used during testing and are not included in the final application build. This aligns with the lab's goal: keep production code clean while giving the test harness everything it needs to run isolated, repeatable unit tests.

PART 2: Project Structure & Test Cases

This demo project is intentionally small, but it's built exactly the way a real Java service would be built. That's what makes the lab useful: we're not just learning Mockito syntax, but learning how real teams isolate logic and test it without depending on external systems.

The project has five main classes under `src/main/java`:

- **StockPosition** — a tiny data class that just holds a stock symbol and quantity. It exists so the service can work with structured input instead of loose variables.
- **StockService** — represents an external API that would normally fetch stock prices. In real life this might call Yahoo Finance or Bloomberg. In the lab, we mock it so we can control the price values.
- **TradeRepository** — stands in for a database write. It's intentionally a void method so we can demonstrate Mockito's behavior verification.
- **AuditService** — another void dependency, used to show how to verify that an audit event was recorded.
- **PortfolioService** — the “real” class under test. It calculates total portfolio value, computes a management fee, handles invalid symbols safely, and calls the repository + audit service. This is the only class that isn't mocked.

There's also **MockitoDemoApp**, which is just a runnable version of the service using simple lambda implementations. Running it once proves the application works before you start mocking anything.

Test Classes and What They Demonstrate

Each test class is intentionally focused on one Mockito concept so students can see each feature in isolation.

- **PortfolioServicePlainMockTest** This is the “hello world” of Mockito. You create mocks, stub return values, and assert the final total. It proves that the service’s math works when you control the prices.
- **PortfolioServiceVerificationTest** This test is all about behavior. Instead of checking return values, you check that the service *called* the right collaborators with the right arguments. This is how you test void methods in real systems.
- **PortfolioServiceExceptionStubbingTest** This test forces the mock to throw an exception so you can verify that the service handles it gracefully. It’s a safe way to test error paths without needing a real failure.
- **PortfolioServiceSpyTest** A spy wraps a real object but lets you override specific methods. This test shows how partial mocking works and why you should use it sparingly as it is powerful, but easy to misuse.
- **PortfolioServiceAnnotationTest** This is the “grown-up” version of Mockito setup. Instead of manually creating mocks, you use `@Mock`, `@InjectMocks`, and `MockitoExtension`. This is how most enterprise teams structure their tests.

PART 3: Test Results

All six tests passed with zero failures, zero errors, and zero skipped tests. The Surefire TXT, XML, and HTML reports confirm that each test class executed successfully and behaved exactly as the lab intended (see Appendix 3). In practical terms, this means the PortfolioService’s business logic works for the scenarios we tested, the service interacts with its collaborators correctly, error handling behaves as expected, and the annotation-based setup integrates cleanly with JUnit 5.

Table 1: Mockito Test Results

Test Class	Concept Proven	Tests Run	Failures	Errors	Result
PortfolioServicePlainMockTest	mock() + when().thenReturn()	1	0	0	✓ Passed
PortfolioServiceVerificationTest	verify(), times(), ArgumentCaptor	2	0	0	✓ Passed
PortfolioServiceExceptionStubbingTest	thenThrow() + safe handling	1	0	0	✓ Passed
PortfolioServiceSpyTest	spy() + doReturn().when()	1	0	0	✓ Passed
PortfolioServiceAnnotationTest	@Mock + @InjectMocks + MockitoExtension	1	0	0	✓ Passed

These results are encouraging, but they only validate the specific behaviors covered by the test suite. Mockito tests intentionally focus on isolated logic, so they show the “happy path,” one controlled failure path, a single spy override, and one annotation setup. They do not reveal untested branches, missed logic, or deeper structural issues. Tools like JaCoCo would show how much of the service logic was actually executed, while SonarQube and CodeMR would highlight duplicated logic, complexity, coupling, and potential maintainability concerns.

Another limitation is performance. Mockito tests run in milliseconds and validate functional behavior, but they do not simulate real traffic, large datasets, or production-level concurrency. In most organizations, load testing is handled by a separate performance engineering team using tools like JMeter or Gatling. These tests answer questions that unit tests cannot: How does the service behave with thousands of positions? Does it slow down under peak traffic? Are there memory leaks or bottlenecks? Unit tests avoid these concerns by design, but in production, external dependencies (APIs, databases, caching layers, AEM dispatcher rules) absolutely affect performance. Mockito proves correctness of the logic, but not scalability or stability under real-world load.

PART 4: Key Takeaways

Running this lab made it clear how valuable isolated unit testing is. Mockito gives us a practical way to test logic without needing a real API, database, or audit system. Stubbing lets us control collaborator behavior, making tests predictable, repeatable, and safe. It also allows us to test failure paths without exposing customer data or triggering real system errors that could lead to CAPAs or RCA investigations.

The lab also highlighted the limits of unit testing. Passing tests only confirm that the behaviors we checked are correct — they don't guarantee full coverage, structural quality, or production-readiness. That's where tools like JaCoCo, SonarQube, and CodeMR become essential. They reveal untested branches, complexity, coupling, and maintainability concerns that unit tests alone cannot detect.

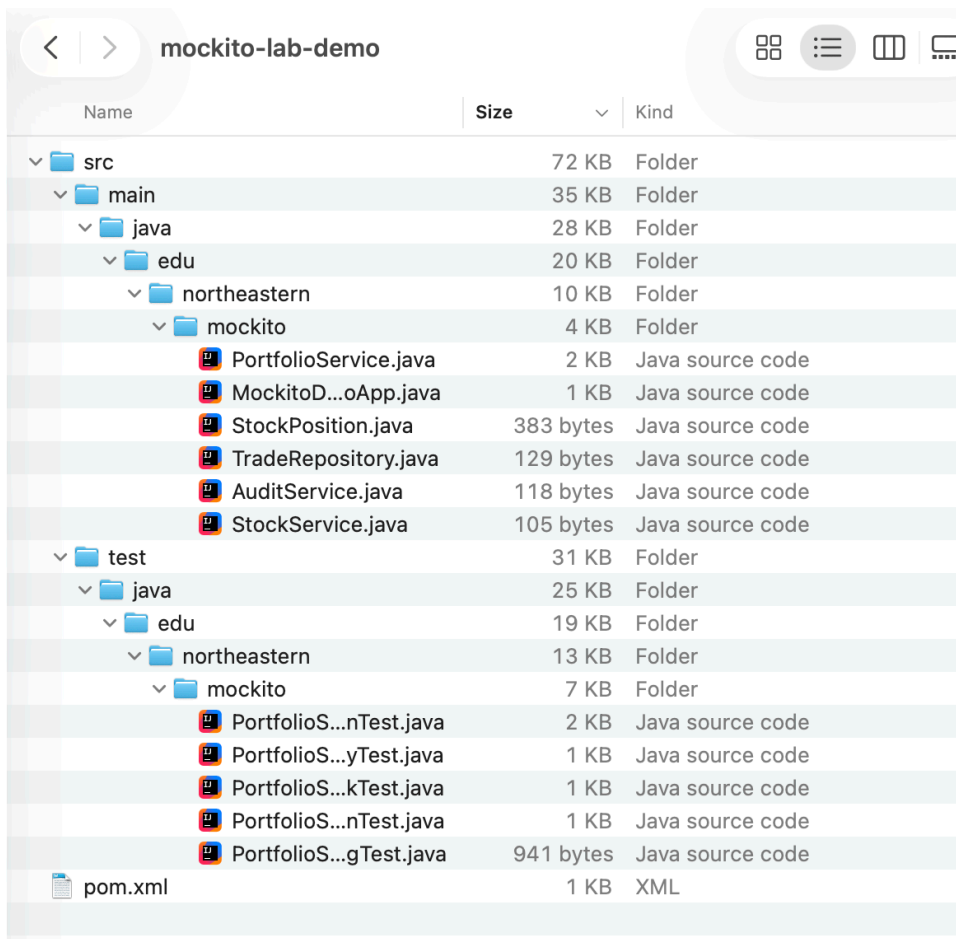
This directly connects to my experience at Abbott. Every team had to manually test their own AEM site whenever shared design-system components changed, which created friction and inconsistent regression coverage. The lab reinforced that a better approach exists: isolate the logic, test it once, and share the results. Mockito can't test AEM pages directly, but the practice of replacing external dependencies with controlled mocks is exactly how AEM's wcm.io testing framework works. Combined with coverage and quality tools, the design system team could have owned automated regression testing for shared components — eliminating cross-functional manual testing and reducing the risk of regressions in production.

APPENDIX 1: Technical Implementation

Figure 1. Java & Maven Verification in Terminal

```
[blackcatranch-m3@JRC-M3 ~ % java -version ]
openjdk version "25.0.3" 2026-04-21 LTS
OpenJDK Runtime Environment Temurin-25.0.3+9 (build 25.0.3+9-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.3+9 (build 25.0.3+9-LTS, mixed mode, sharing)
[blackcatranch-m3@JRC-M3 ~ % mvn -version ]
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/homebrew/Cellar/maven/3.9.16/libexec
Java version: 25.0.3, vendor: Eclipse Adoptium, runtime: /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home
Default locale: en_US, platform encoding: UTF-8
OS name: "mac os x", version: "26.5", arch: "aarch64", family: "mac"
blackcatranch-m3@JRC-M3 ~ %
```

Figure 2. Confirming setup of mockito-lab-demo



The screenshot shows a file explorer window titled 'mockito-lab-demo'. The interface includes a sidebar with navigation icons, a main table of files and folders, and a top bar with view toggles. The table has columns for 'Name', 'Size', and 'Kind'. The directory structure is as follows:

Name	Size	Kind
src	72 KB	Folder
main	35 KB	Folder
java	28 KB	Folder
edu	20 KB	Folder
northeastern	10 KB	Folder
mockito	4 KB	Folder
PortfolioService.java	2 KB	Java source code
MockitoD...oApp.java	1 KB	Java source code
StockPosition.java	383 bytes	Java source code
TradeRepository.java	129 bytes	Java source code
AuditService.java	118 bytes	Java source code
StockService.java	105 bytes	Java source code
test	31 KB	Folder
java	25 KB	Folder
edu	19 KB	Folder
northeastern	13 KB	Folder
mockito	7 KB	Folder
PortfolioS...nTest.java	2 KB	Java source code
PortfolioS...yTest.java	1 KB	Java source code
PortfolioS...kTest.java	1 KB	Java source code
PortfolioS...nTest.java	1 KB	Java source code
PortfolioS...gTest.java	941 bytes	Java source code
pom.xml	1 KB	XML

Figure 3. Mvn test results

```
[INFO] Running edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -
- in edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.013 s -
- in edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -
- in edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.038 s -
- in edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.567 s
[INFO] Finished at: 2026-07-16T13:39:40-07:00
[INFO]
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 4. Mvn clean compile results

```
● ● ● mockito-lab-demo -- -zsh -- 80x24

[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ mockito-lab-demo ---
[INFO] Deleting /Users/blackcatranch-m3/Documents/mockito-lab-demo/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/moc
kito-lab-demo/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 6 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot
run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets
the location of system modules automatically
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.532 s
[INFO] Finished at: 2026-07-16T13:40:33-07:00
[INFO]
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 5. Mvn clean test results

```
● ● ● mockito-lab-demo -- -zsh -- 80x24

[INFO] Running edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -
- in edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.013 s -
- in edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -
- in edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.038 s -
- in edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.590 s
[INFO] Finished at: 2026-07-16T13:41:09-07:00
[INFO]
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

APPENDIX 2: Mvn test results

Figure 6. Mvn clean compile results

```
- in edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.013 s -
- in edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -
- in edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Running edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.038 s -
- in edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.590 s
[INFO] Finished at: 2026-07-16T13:41:09-07:00
[INFO] -----
[blackcatranch-m3@JRC-M3 mockito-lab-demo % mvn clean compile ]
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ mockito-lab-demo ---
[INFO] Deleting /Users/blackcatranch-m3/Documents/mockito-lab-demo/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/moc
kito-lab-demo/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 6 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot
run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets
the location of system modules automatically
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.513 s
[INFO] Finished at: 2026-07-16T14:49:56-07:00
[INFO] -----
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 7. java -cp target/classes edu.northeastern.mockito.MockitoDemoApp results

```
[blackcatranch-m3@JRC-M3 mockito-lab-demo % java -cp target/classes edu.northeastern.mockito.MockitoDemoApp
Portfolio total value: $4256.25
Management fee: $42.56
Saved valuation for student: $4256.25
Audit event: VALUATION_SAVED for student
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 8. Run individual test classes results

```
blackcatranch-m3@JRC-M3 mockito-lab-demo % mvn test -Dtest=PortfolioServicePlainMockTest
mvn test -Dtest=PortfolioServiceVerificationTest
mvn test -Dtest=PortfolioServiceExceptionStubbingTest
mvn test -Dtest=PortfolioServiceSpyTest
mvn test -Dtest=PortfolioServiceAnnotationTest
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 5 source files with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
[ not setting the location of system modules may lead to class files that cannot run on JDK 17 ]
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules
automatically
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServicePlainMockTest
[Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://jav
adoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath
has been appended
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-bu
ddy-agent/1.17.7/byte-buddy-agent-1.17.7.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warnin
g
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more informati
on
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.399 s -- in edu.northeastern.mockito.P
ortfolioServicePlainMockTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.302 s
[INFO] Finished at: 2026-07-16T14:53:00-07:00
[INFO] -----
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resource
s
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resource
s
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServiceVerificationTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future release
s of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://jav
adoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org.mockito/Mockito.html#0.3
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath
has been appended
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-bu
ddy-agent/1.17.7/byte-buddy-agent-1.17.7.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warnin
g
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more informati
on
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.389 s -- in edu.northeastern.mockito.P
ortfolioServiceVerificationTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.914 s
[INFO] Finished at: 2026-07-16T14:53:01-07:00
[INFO] -----
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resource
s
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resource
s
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://jav
adoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-bu
ddy-agent/1.17.7/byte-buddy-agent-1.17.7.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warnin
g
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more informati
on
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath
has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.368 s -- in edu.northeastern.mockito.P
ortfolioServiceExceptionStubbingTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.873 s
[INFO] Finished at: 2026-07-16T14:53:03-07:00
[INFO] -----
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resource
s
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resource
s
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServiceSpyTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as
an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mock
ito.html#0.3
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-buddy-agent/1.17.7/byte-buddy-agent-1.
17.7.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.409 s -- in edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.925 s
[INFO] Finished at: 2026-07-16T14:53:04-07:00
[INFO]
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern.mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServiceAnnotationTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as
an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mock
ito.html#0.3
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-buddy-agent/1.17.7/byte-buddy-agent-1.
17.7.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.377 s -- in edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.902 s
[INFO] Finished at: 2026-07-16T14:53:06-07:00
[INFO]
[INFO] -----
blackcatranch-m3@JRC-M3 mockito-lab-demo %

```

Figure 9. Run the full suite – mvn clean test

```
blackcatranch-m3@JRC-M3 mockito-lab-demo % mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.northeastern:mockito-lab-demo >-----
[INFO] Building mockito-lab-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ mockito-lab-demo ---
[INFO] Deleting /Users/blackcatranch-m3/Documents/mockito-lab-demo/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 6 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 5 source files with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.northeastern.mockito.PortfolioServicePlainMockTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future
nt to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap cl
WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/bytebuddy-agent-1.14.1.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more in
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.437 s -- in edu.northeastern.mc
[INFO] Running edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -- in edu.northeastern.mc
[INFO] Running edu.northeastern.mockito.PortfolioServiceAnnotationTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s -- in edu.northeastern.mc
[INFO] Running edu.northeastern.mockito.PortfolioServiceVerificationTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.009 s -- in edu.northeastern.mc
[INFO] Running edu.northeastern.mockito.PortfolioServiceSpyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.031 s -- in edu.northeastern.mc
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.571 s
[INFO] Finished at: 2026-07-16T15:02:43-07:00
[INFO] -----
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 10. Generate HTML Surefire report with mvn clean test surefire-report:report

```
[INFO] >>> surefire-report:3.6.0-M1:report (default-cli) > [surefire]test @ mockito-lab-demo >>>
[INFO] --- resources:3.4.0:resources (default-resources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/main/resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ mockito-lab-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ mockito-lab-demo ---
[INFO] skip non existing resourceDirectory /Users/blackcatranch-m3/Documents/mockito-lab-demo/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-lab-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 5 source files with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automat
ally
[INFO] --- surefire:3.5.6:test (default-test) @ mockito-lab-demo ---
[INFO] Skipping execution of surefire because it has already been run for this configuration
[INFO] <<< surefire-report:3.6.0-M1:report (default-cli) < [surefire]test @ mockito-lab-demo <<<
[INFO] --- surefire-report:3.6.0-M1:report (default-cli) @ mockito-lab-demo ---
[INFO] Rendering content with org.apache.maven.skis:maven-fluido-skin:jar:2.0.0-M9 skin
[WARNING] Unable to locate Test Source XRef to link to -- DISABLED
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.055 s
[INFO] Finished at: 2026-07-16T15:07:24-07:00
[INFO] -----
lackcatranch-m3@JRC-M3 mockito-lab-demo %
```

Figure 11. Find and open generated HTML report on MacOS

```
blackcatranch-m3@JRC-M3 mockito-lab-demo % find target -type f -name "*.html"

target/reports/surefire.html
blackcatranch-m3@JRC-M3 mockito-lab-demo % open target/reports/surefire.html
blackcatranch-m3@JRC-M3 mockito-lab-demo %
```

APPENDIX 3: Evidence of 3 types of generated reports

Figure 12. Surefire Generated HTML Report

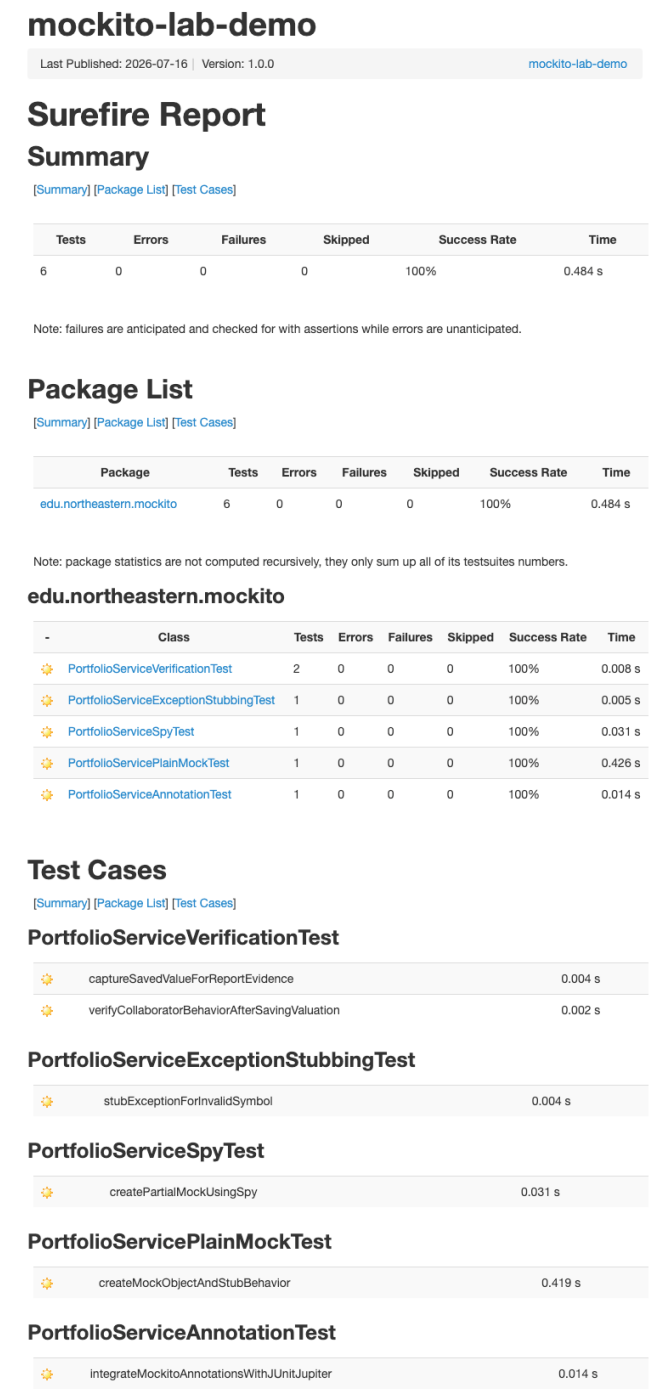


Figure 13. Evidence of Generated Text & XML Reports From Finder

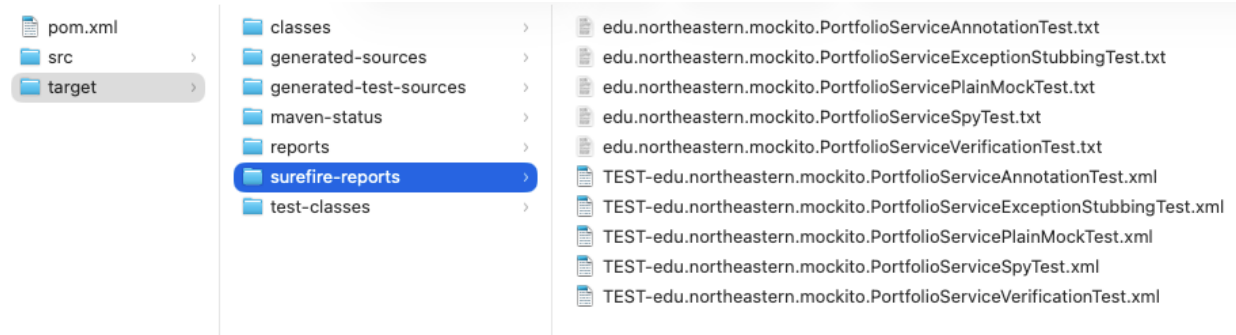


Figure 14. Generated Text Reports



Figure 15. Generated XML Report Example

```

st.xml TEST-edu.northeastern.mockito.PortfolioServiceExceptionStubbingTest.xml TEST-edu.northeastern.mockito.PortfolioServicePlainMockTest.xml
Users > blackcatranch-m3 > Documents > mockito-lab-demo > target > surefire-reports > TEST-edu.northeastern.mockito.PortfolioServicePlainMockTest.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2 <testsuite xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="https://maven.apache.org/surefire/maven-surefire-
3 <properties>
4 <property name="java.specification.version" value="25"/>
5 <property name="sun.jnu.encoding" value="UTF-8"/>
6 <property name="java.class.path" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo/target/test-classes:/Users/blackcatranch-m3/Docu
7 <property name="java.vm.vendor" value="Eclipse Adoptium"/>
8 <property name="sun.arch.data.model" value="64"/>
9 <property name="java.vendor.url" value="https://adoptium.net/">
10 <property name="user.timezone" value="America/Los_Angeles"/>
11 <property name="os.name" value="Mac OS X"/>
12 <property name="java.vm.specification.version" value="25"/>
13 <property name="sun.java.launcher" value="SUN_STANDARD"/>
14 <property name="user.country" value="US"/>
15 <property name="sun.boot.library.path" value="/Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/lib"/>
16 <property name="sun.java.command" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo/target/surefire/surefirebooter-20260716150723438
17 <property name="http.nonProxyHosts" value="local|*.local|169.254.16|*.169.254.16"/>
18 <property name="jdk.debug" value="release"/>
19 <property name="surefire.test.class.path" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo/target/test-classes:/Users/blackcatranch-
20 <property name="sun.cpu.endian" value="little"/>
21 <property name="user.home" value="/Users/blackcatranch-m3"/>
22 <property name="user.language" value="en"/>
23 <property name="java.specification.vendor" value="Oracle Corporation"/>
24 <property name="java.version.date" value="2026-04-21"/>
25 <property name="java.home" value="/Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home"/>
26 <property name="file.separator" value="/">
27 <property name="basedir" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo"/>
28 <property name="java.vm.compressedOopsMode" value="Zero based"/>
29 <property name="line.separator" value="
30 <property name="java.vm.specification.vendor" value="Oracle Corporation"/>
31 <property name="java.specification.name" value="Java Platform API Specification"/>
32 <property name="apple.awt.application.name" value="ForkedBooter"/>
33 <property name="surefire.real.class.path" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo/target/surefire/surefirebooter-2026071615
34 <property name="sun.management.compiler" value="HotSpot 64-Bit Tiered Compilers"/>
35 <property name="ftp.nonProxyHosts" value="local|*.local|169.254.16|*.169.254.16"/>
36 <property name="java.runtime.version" value="25.0.3+9-LTS"/>
37 <property name="user.name" value="blackcatranch-m3"/>
38 <property name="stdout.encoding" value="UTF-8"/>
39 <property name="path.separator" value=":">
40 <property name="os.version" value="26.5"/>
41 <property name="java.runtime.name" value="OpenJDK Runtime Environment"/>
42 <property name="file.encoding" value="UTF-8"/>
43 <property name="java.vm.name" value="OpenJDK 64-Bit Server VM"/>
44 <property name="java.vendor.version" value="Temurin-25.0.3+9"/>
45 <property name="localRepository" value="/Users/blackcatranch-m3/.m2/repository"/>
46 <property name="java.vendor.url.bug" value="https://github.com/adoptium/adoptium-support/issues"/>
47 <property name="java.io.tmpdir" value="/var/folders/81/nt8dq7n94pbckhstyrbnpq_c0000gn/T"/>
48 <property name="java.version" value="25.0.3"/>
49 <property name="user.dir" value="/Users/blackcatranch-m3/Documents/mockito-lab-demo"/>
50 <property name="os.arch" value="aarch64"/>
51 <property name="java.vm.specification.name" value="Java Virtual Machine Specification"/>
52 <property name="native.encoding" value="UTF-8"/>
53 <property name="java.library.path" value="/Users/blackcatranch-m3/Library/Java/Extensions:/Library/Java/Extensions:/Network/Library/Java/Ext
54 <property name="java.vm.info" value="mixed mode, sharing"/>
55 <property name="stderr.encoding" value="UTF-8"/>
56 <property name="java.vendor" value="Eclipse Adoptium"/>
57 <property name="java.vm.version" value="25.0.3+9-LTS"/>
58 <property name="stdin.encoding" value="UTF-8"/>
59 <property name="sun.io.unicode.encoding" value="UnicodeBig"/>
60 <property name="socksNonProxyHosts" value="local|*.local|169.254.16|*.169.254.16"/>
61 <property name="java.class.version" value="69.0"/>
62 </properties>
63 <testcase name="createMockObjectAndStubBehavior" classname="edu.northeastern.mockito.PortfolioServicePlainMockTest" time="0.419">
64 <system-err><![CDATA[Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the
65 WARNING: A Java agent has been loaded dynamically (/Users/blackcatranch-m3/.m2/repository/net/bytebuddy/byte-buddy-agent/1.17.7/byte-buddy-agent-
66 WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
67 WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
68 WARNING: Dynamic loading of agents will be disallowed by default in a future release
69 ]]></system-err>
70 </testcase>
71 </testsuite>

```

REFERENCES:

Mockito. (n.d.). *Mockito framework* (Version 5.23.0) [Software documentation].
<https://javadoc.io/doc/org.mockito/mockito-core/latest/index.html>

JUnit Team. (n.d.). *JUnit 5 user guide*. <https://junit.org/junit5/docs/current/user-guide/>

Apache Maven Project. (n.d.). *Maven Surefire plugin*. <https://maven.apache.org/surefire/maven-surefire-plugin/>

Methodology and Use of AI Tools

I relied on the coursework “Mockito Lab End to End Deep Dive PDF” guide, Class Connect and other Canvas & Coursera materials for CSYE6255.

The analysis, observations, and conclusions reflect the author's own hands-on experience with the tool and tasks analyzed. All content has been reviewed for accuracy and represents the author's understanding of the results.

Reference to Claude LLM – Extraction of demo code to to local:
<https://claude.ai/share/61d09a48-6c4d-4afa-9487-3e2e1bcfed09>

Reference to Claude LLM conversation on draft and rubric compliance:
<https://claude.ai/share/2babf135-3c6a-4ce7-b2c5-8cfdb0b14f6f>